

Dr. Thejus Mahajan
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Ms. Anna Sacher
Institute of Computational Biology
Helmholtz Munich
Ingolstädter Landstraße 1
85764 Neuherberg

Staff Scientist (f/m/x) in Computational Biology (Job ID: 102868)

Dear Ms. Sacher,

Bridging rigorous physical science computing with modern biomedical workflows has defined my research career. The opportunity to shape scientific strategy and elevate methodological standards at the Computational Health Center aligns perfectly with my drive to engineer reproducible single-cell pipelines. My transition from a Ph.D. in computational systems modeling to bioinformatics pipeline and ML engineering makes me a highly suitable candidate to drive your single-cell and computational biology projects.

My career has been defined by bridging rigorous physical science computing with modern biomedical workflows, and I have a proven track record of mentoring and coordinating interdisciplinary teams. Throughout my post-doc and recent roles, I have spearheaded complex research designs while guiding junior colleagues and students in advanced computational techniques. At my current internship with HealthTwist GmbH, I took ownership of code quality for Germany's interventional radiology quality registry. I externalized hardcoded logic, verified byte-level reproducibility, and reduced a 1,834-line production pipeline by 26%—establishing the exact kind of reproducible workflows and shared best practices you require.

Your need for expertise in modern scientific computing frameworks (JAX, PyTorch, scikit-learn) aligns perfectly with my core technical strengths:

- **Scientific Computing & AI (JAX/PyTorch/scikit-learn):** During my post-doc, I led the translation of a legacy Fortran model into a high-performance Python engine using **Google JAX**. By engineering branchless logic and vectorized operations, I achieved direct compatibility with **Google TPUs and GPUs** for massive parallelization. Because JAX shares core paradigms with **PyTorch**, I possess the deep, under-the-hood expertise required to scale AI architectures. I am also actively expanding my applied ML capabilities with scikit-learn and Tidymodels architectures.
- **Mentorship & Scientific Strategy:** I have extensive experience coordinating research initiatives and preparing scientific content. From coaching state-level teams as an educator to spearheading evolutionary systems modeling at Helmholtz Hereon, I excel at translating complex quantitative methods to diverse, interdisciplinary teams.
- **Reproducible Workflows & Single-Cell Concepts:** Highly proficient in building reproducible, containerized workflows (Nextflow/DSL2) and processing multi-omic data. I bring a strong understanding of the statistical modeling required to extract signal from high-dimensional biological datasets like transcriptomics.

I am deeply motivated by Helmholtz Munich's mission to bridge fundamental methodological development with translational biomedicine. My robust mathematical background and dedication to reproducible engineering will allow me to elevate your lab's computational standards while mentoring the next generation of researchers. I welcome the opportunity to discuss further and am available for an interview.

Sincerely



Dr. Thejus Mahajan